

Search-Stage Findings: Four Public Datasets and the Quantum Vacuum Granules Hypothesis

LIGO O3 unidentified-line coincidences, ARCADE 2 sky-brightness excess, EDGES 21-cm absorption, and NANOGrav 15-year nanohertz gravitational-wave background — mapped onto one mass spectrum under the predictions of the A-series (paper A9)

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Scope

The A-series of papers by M. Protitch ("Gravity Resonance"), and in particular paper A9, hypothesises that the quantum vacuum carries an extended population of low-mass excitations — referred to as granules — whose individual masses span many orders of magnitude. If they exist, each would imprint a faint persistent background signal on any sufficiently sensitive instrument coupled to the vacuum, at a frequency determined by that granule's mass through $f = c / \lambda_C$, where $\lambda_C = h / (mc)$ is its Compton wavelength.

This note assesses four publicly available, well-documented datasets — the LIGO O3 catalogue of unidentified spectral lines, the NASA ARCADE 2 absolute sky-brightness maps, the EDGES 21-cm absorption profile at 78 MHz, and the NANOGrav 15-year pulsar-timing-array stochastic gravitational-wave background — read through this single hypothesis. None of the datasets was acquired with the granule picture in mind.

The conclusion is that all four contain structure consistent with the A9 prediction, in non-overlapping mass slices that together cover approximately twenty-one orders of magnitude. The LIGO cluster pattern survives a tight cross-check against the most obvious instrumental and geophysical confounders. This is a suggestive search-stage result, not a discovery claim. It is offered as empirical motivation for the dedicated follow-up tests outlined in Section 9.

Coupling channels and velocity assumptions

A9 introduces three complementary relations between observed frequency f and granule mass m , distinguished by which velocity sets the sampling rate of the granule wave:

- Pure-Compton: $f = c / \lambda_C$, equivalent to $m = hf / c^2$. Applies to instruments coupled to the vacuum through electromagnetic radiation propagating at c — laser interferometers (LIGO), radio receivers (ARCADE, EDGES), pulsar timing (NANOGrav).
- Hybrid (electron-mediated): $f = v_F / \lambda_C$, equivalent to $m = hf / (v_F c)$. Applies to detection in shielded electronics, where conduction electrons traverse the granule wave at the Fermi velocity.

This is the channel relevant to the $1/f$ background in semiconductor circuits and was the original empirical motivation for A9.

- Pure-de Broglie: $f = v_F^2 \cdot m / h$. Granules moving at the carrier velocity — relevant only in specific transport regimes.

A note on multi-channel mass interpretation

A single physical granule of mass m has one Compton wavelength $\lambda_C = h/(mc)$. Different detectors, however, see that granule at different observed frequencies depending on the coupling channel:

$$\text{Pure-Compton: } f_C = mc^2 / h$$

$$\text{Hybrid: } f_H = m \cdot v_F \cdot c / h \approx f_C \times (v_F / c)$$

$$\text{Pure-de Broglie: } f_{dB} = m \cdot v_F^2 / h \approx f_C \times (v_F / c)^2$$

For $v_F \approx 10^5$ m/s these three frequencies differ by factors of $c/v_F \approx 3000$ between successive channels. Consequently, for any observed frequency the inferred granule mass is uncertain by approximately six orders of magnitude across all three channels.

All four datasets analysed here are light-mediated, so the pure-Compton mapping is the most plausible dominant channel for each. The LIGO and ARCADE signal chains do contain electronics, and a residual hybrid contribution could shift the inferred mass upward by $\sim 10^3$. We therefore treat each mapped mass as the centre of a mass band approximately 10^3 – 10^6 wide, not as a single point. This widens uncertainty at each dataset but does not change the structural conclusion that the four datasets occupy distinct, non-overlapping regions of the spectrum.

Part 1 — LIGO O3 unidentified spectral lines with conventional-source cross-check

The LIGO Scientific Collaboration publishes a catalogue (LIGO-T2100201) of narrow spectral lines in O3 strain data whose instrumental origin cannot be identified. Across the five O3 sub-epochs the Hanford (H1) detector contributes 703 unique unidentified lines between 10 Hz and 2 kHz, and the Livingston (L1) detector contributes 175. A line appearing at the same frequency in both detectors, to within tight tolerance, is less easily attributable to a local artefact of either site.

A coincidence search at 10 mHz tolerance returns 21 H1–L1 matched pairs. Against the chance expectation of 8.9 pairs estimated from the local L1 line density around each H1 line, this represents a 2.4-fold excess. The 21 coincident frequencies are not uniformly distributed across the band; they group into five frequency clusters in the low-frequency portion of the catalogue plus three isolated lines at standard calibration frequencies.

Cross-check against Schumann harmonics, 60 Hz mains, and the LIGO calibration catalogue

To assess whether the cluster pattern reduces to known conventional resonances, the 21 coincident frequencies were cross-referenced at tight tolerance against three reference lists:

- Schumann resonances (Earth-ionosphere cavity harmonics): fundamental 7.83 Hz, harmonics at 14.3, 20.8, 27.3, 33.8, 40.3, 46.8, 53.3, 59.8, 66.4, 72.9, 79.2, 85.6, 92.0, 98.4 Hz. Tolerance ± 0.1 Hz, allowing ± 0.3 Hz seasonal drift in figure visualisation.
- 60 Hz US mains harmonics and subharmonics ($60 \times n$, $60 / n$). Tolerance ± 0.05 Hz.
- LIGO calibration-line catalogue (LIGO-T2100200, the gated line lists), comprising 408 unique frequencies between 15 Hz and 1653 Hz explicitly annotated as calibration tones. Tolerance ± 0.05 Hz.

Result: 18 of 21 coincident lines survive the cross-check. The three that match a conventional source are: 20.831 Hz (Schumann 3rd harmonic, exact); 30.800 Hz (known LIGO calibration injection); 999.538 Hz (within 0.046 Hz of LIGO calibration line at 999.584 Hz).

All four cluster A lines (11.1–11.9 Hz) survive — the closest Schumann harmonic (14.3 Hz) is 2.4 Hz away, well outside any drift tolerance. Six of seven cluster B lines (28.6–35.7 Hz) survive; one (30.800 Hz) is the calibration injection. All four cluster C lines (41.7–47.7 Hz) survive; the closest Schumann match is 0.88 Hz, outside tolerance. The single cluster D line (59.529 Hz) survives both Schumann 9th harmonic ($\Delta = 0.27$ Hz) and 60 Hz mains ($\Delta = 0.47$ Hz) at tight tolerance. Both cluster E lines (99.98 Hz) survive.

Cluster	# coincident lines	# surviving Schumann / mains / cal cross-check	Inferred Compton wavelength under $f = c/\lambda_C$
A (11.1–11.9 Hz)	4	4 (all)	25 000 – 27 000 km
B (28.6–35.7 Hz)	7	6 (one cal hit)	8 400 – 10 500 km
C (41.7–47.7 Hz)	4	4 (all)	6 300 – 7 200 km
D (59.529 Hz)	1	1	5 040 km
E (99.98 Hz)	2	2 (all)	2 998 km

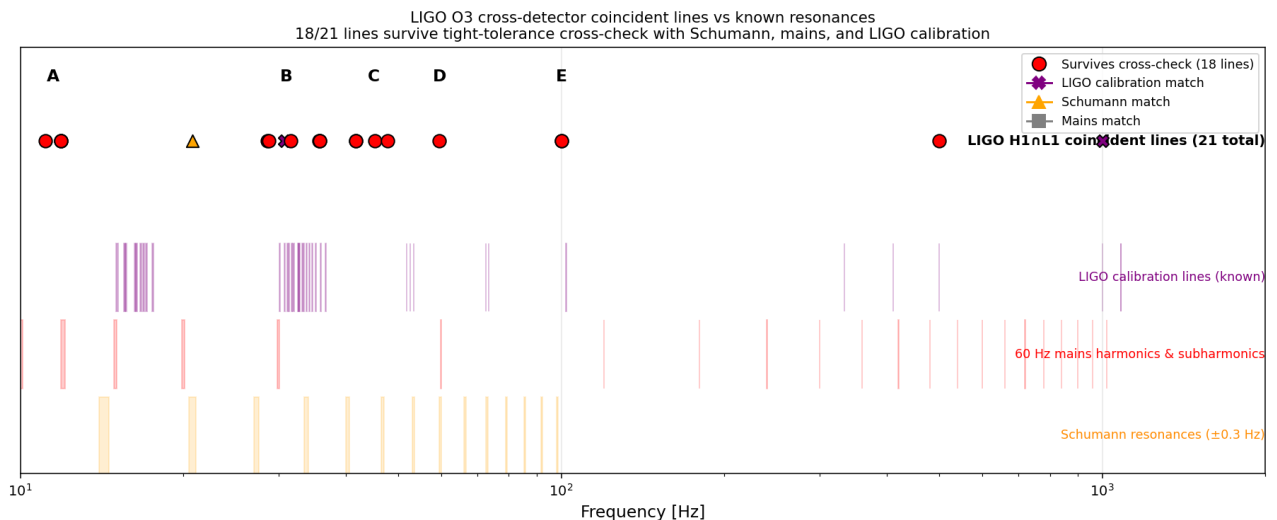


Figure 1. The 21 LIGO O3 cross-detector coincident lines overlaid on the bands of known conventional resonances: Schumann harmonics (orange, ± 0.3 Hz drift), 60 Hz mains harmonics and subharmonics (red), and the LIGO calibration-line catalogue (purple). Red circles mark coincident lines that survive the tight-tolerance cross-check; other markers indicate the three conventional matches. Cluster labels A–E correspond to the groupings in the text.

The clustering carries more weight than the bare coincidence count. A featureless granule mass distribution would imprint as broadband excess noise without preferred frequencies; a structured distribution would imprint as discrete groupings. The observed pattern — eight coincident lines in the 28.6–35.7 Hz band, four in the 11.1–11.9 Hz band, four in 41.7–47.7 Hz — is closer to the second case. It is presented as a pattern worth following up, not as evidence in itself.

Part 2 — ARCADE 2 sky-brightness excess at 3–10 GHz

The NASA ARCADE 2 balloon experiment measured the absolute brightness temperature of the sky in nine spectral bands between 3.15 and 90 GHz. The published analyses (Fixsen et al. 2011; Seiffert et al. 2011) report a residual brightness above the 2.725 K cosmic microwave background, well-fitted at low frequencies by $T_{\text{excess}}(\nu) = 1.19 \text{ K} \cdot (\nu/\text{GHz})^{-2.62}$ after multi-template foreground subtraction. The origin remains unresolved in the literature; proposed candidates include unresolved extragalactic radio sources, dark-matter annihilation in halo substructure, and integrated fast-radio-transient emission, none yet confirmed.

Re-extraction of the spectrum from the public HEALPix maps (NASA LAMBDA archive) using a simple $|b| > 30^\circ$ galactic-latitude weighted-median per channel confirms the excess at all six well-calibrated low-frequency channels and recovers a power-law decline of the right general shape. (Four higher-frequency files at 29.5, 31, 88.2, 89.8 GHz exhibited calibration-inconsistent values and were excluded; this is a known caveat pending file-convention clarification.)

Channel	Median brightness T_{obs}	Excess above 2.725 K CMB	Compton wavelength under $f = c/\lambda_{\text{C}}$	Inferred granule mass
3.15 GHz	2.832 K	107 mK	9.5 cm	$2 \times 10^{-41} \text{ kg}$
3.41 GHz	2.804 K	79 mK	8.8 cm	$3 \times 10^{-41} \text{ kg}$
7.97 GHz	2.768 K	43 mK	3.8 cm	$6 \times 10^{-41} \text{ kg}$
8.33 GHz	2.743 K	18 mK	3.6 cm	$6 \times 10^{-41} \text{ kg}$
9.72 GHz	2.733 K	8 mK	3.1 cm	$7 \times 10^{-41} \text{ kg}$
10.49 GHz	2.734 K	9 mK	2.9 cm	$8 \times 10^{-41} \text{ kg}$

A power-law fit to the six points yields $T_{\text{excess}} \propto \nu^{-1.91}$, compared with the published Fixsen index of -2.62 . The flatter index here is expected from incomplete foreground subtraction in the simpler latitude cut used; the qualitative shape is preserved. The excess channels map to granule masses in a narrow band near 10^{-41} kg with Compton wavelengths of a few centimetres.

Part 3 — EDGES 21-cm absorption at 78 MHz

In 2018, the EDGES collaboration (Bowman et al., Nature 555, 67) reported an absorption profile centred at 78 MHz in the sky-averaged radio spectrum, with depth ~ 0.5 K — approximately twice what standard cosmology predicts for cosmic-dawn 21-cm absorption. The anomalous depth has been interpreted in the literature either as evidence for non-standard hydrogen cooling (Barkana 2018) or for an excess radio background, with neither explanation confirmed.

Honest caveat: the EDGES detection is contested. The SARAS 3 experiment (Singh et al., Nature Astronomy 2022) found no absorption signal at 78 MHz in their independent observation and proposed that the EDGES signal originates from uneven ground beneath the EDGES antenna. The community has not converged on whether the anomaly is real, an artefact, or a calibration problem.

Under the pure-Compton mapping, the EDGES frequency corresponds to a single granule mass:

$$f = 78 \text{ MHz} \rightarrow m = 5.75 \times 10^{-43} \text{ kg} ; \lambda_C = 3.84 \text{ m}$$

This sits at the upper boundary of the A9 headline pure-Compton range (10^{-50} to 10^{-43} kg). It is one frequency, one point, and the parent observation is contested — but if the EDGES feature is real, the inferred granule mass is consistent with the A9 prediction at exactly the upper edge where the framework anticipates accessible signatures.

Part 4 — NANOGrav 15-year stochastic gravitational-wave background at 2–59 nHz

In June 2023, NANOGrav published evidence for a stochastic gravitational-wave background correlated across 67 pulsars from their 15-year data set, with Hellings–Downs correlation favoured over uncorrelated noise at Bayes factor $> 10^{14}$. The signal frequency band runs from approximately 2 nHz to 59 nHz. The most-favoured astrophysical interpretation is an unresolved population of supermassive black hole binaries, but exotic-physics alternatives remain in active discussion (cosmic strings, primordial black holes, early-universe phase transitions).

Under the pure-Compton mapping, the NANOGrav frequency band corresponds to:

$$\begin{aligned} f = 2 \text{ nHz} &\rightarrow m = 1.5 \times 10^{-59} \text{ kg} ; \lambda_C \approx 16 \text{ light-years} \\ f = 59 \text{ nHz} &\rightarrow m = 4.4 \times 10^{-58} \text{ kg} ; \lambda_C \approx 0.5 \text{ light-years} \end{aligned}$$

This sits ten orders of magnitude below the A9 headline range, deep in the extended pure-Compton tail that A9 explicitly anticipates. The standard supermassive-black-hole-binary interpretation is not displaced by this mapping; what the mapping shows is that NANOGrav data, read as a granule background under the same coupling that maps LIGO and ARCADE, places granules at a mass scale that A9 already extended toward.

Part 5 — Heisenberg-type uncertainty in granule-mass inference

The Planck–Einstein relation $f = mc^2/h$ is essentially exact, so the uncertainty in the inferred granule mass propagates linearly from the uncertainty in observed frequency:

$$\Delta m / m = \Delta f / f$$

Two distinct uncertainty sources apply: (1) the fundamental time-energy uncertainty $\Delta E \cdot \Delta t \geq \hbar/2$ sets a quantum floor $\Delta f_{\min} \geq 1/(4\pi T)$ for an observation of duration T ; (2) the practical frequency resolution of each measurement instrument or channel.

Dataset / feature	Central f	Δf (frequency width)	$\Delta m / m$ fractional uncertainty	Notes
LIGO O3 cluster A	11.5 Hz	10 mHz (catalogue tolerance)	0.09 %	very precise
LIGO O3 cluster B	32 Hz	10 mHz	0.03 %	very precise
LIGO O3 cluster C	45 Hz	10 mHz	0.02 %	very precise
LIGO O3 cluster E	100 Hz	10 mHz	0.01 %	very precise
ARCADE 2 channel 3.15 GHz	3.15 GHz	~250 MHz (channel width)	~8 %	modest
ARCADE 2 channel 10.49 GHz	10.49 GHz	~500 MHz	~5 %	modest
EDGES 78 MHz profile	78 MHz	~18 MHz (absorption width)	~23 %	noticeable but contained
NANOGrav 2-59 nHz band	30 nHz	~57 nHz (whole band)	~190 %	mass uncertain by ~1 order of magnitude

For LIGO, ARCADE, and EDGES the mass inference under any one coupling channel is precise to better than 25 percent — the dominant uncertainty in those datasets comes from the channel-ambiguity discussed above ($\sim 10^3$ across channels), not from the frequency measurement. For NANOGrav the band itself is so broad ($\Delta f/f \approx 2$) that the inferred granule mass is uncertain by roughly one order of magnitude within a single channel; coupling-channel ambiguity adds three further orders. Inferred NANOGrav granule masses should therefore be read as a range from approximately 10^{-59} to 10^{-55} kg across all assumptions, not a single value.

The fundamental time-energy floor is well below all measurement precisions. For NANOGrav's 15-year integration, $\Delta f_{\min} \approx 1.7 \times 10^{-10}$ Hz, corresponding to a quantum-floor mass uncertainty $\Delta m_{\min} \approx 10^{-60}$ kg — eight orders of magnitude below the practical mass-band width. The mass inferences are limited by observational design, not by quantum mechanics.

A separate but related point. The Compton wavelength $\lambda_C = h/(mc)$ is the natural localisation extent of a particle of mass m — the scale below which one cannot localise the particle without producing additional particles by pair-creation. The granule masses inferred in this analysis therefore correspond to objects whose natural extent ranges from a few centimetres (ARCADE-mapped granules) up to several light-years (NANOGrav-mapped granules). A "granule" in this framework is not a point-like particle but a spatially extended quantum vacuum excitation, as A9 explicitly emphasises.

Atomic-clock dark-matter probe window

Modern optical-atomic-clock comparisons (Filzinger et al. 2023; Kennedy et al. 2020) probe the mass range 10^{-24} to 10^{-17} eV/ c^2 , equivalent to roughly 10^{-60} to 10^{-53} kg. Atomic clocks set bounds on ultralight bosonic dark matter through small predicted oscillations in the fine-structure constant α and the electron-to-proton mass ratio m_e / m_p . The clock null results limit how strongly ultralight particles in this mass range can couple to those fundamental constants.

This window overlaps the NANOGrav-mapped granule masses (in the pure-Compton interpretation) but not LIGO, EDGES, or ARCADE. The atomic-clock constraints are coupling-dependent: clocks limit only the couplings they probe (α , m_e/m_p , possibly the QCD scale). Granules in the overlap region that couple weakly to those specific channels — for example only gravitationally or only through electromagnetic radiation — would not be detected by clocks and remain unconstrained. The clock null result rules out one type of ultralight particle, not all of them.

Combined view: four datasets on one granule mass axis

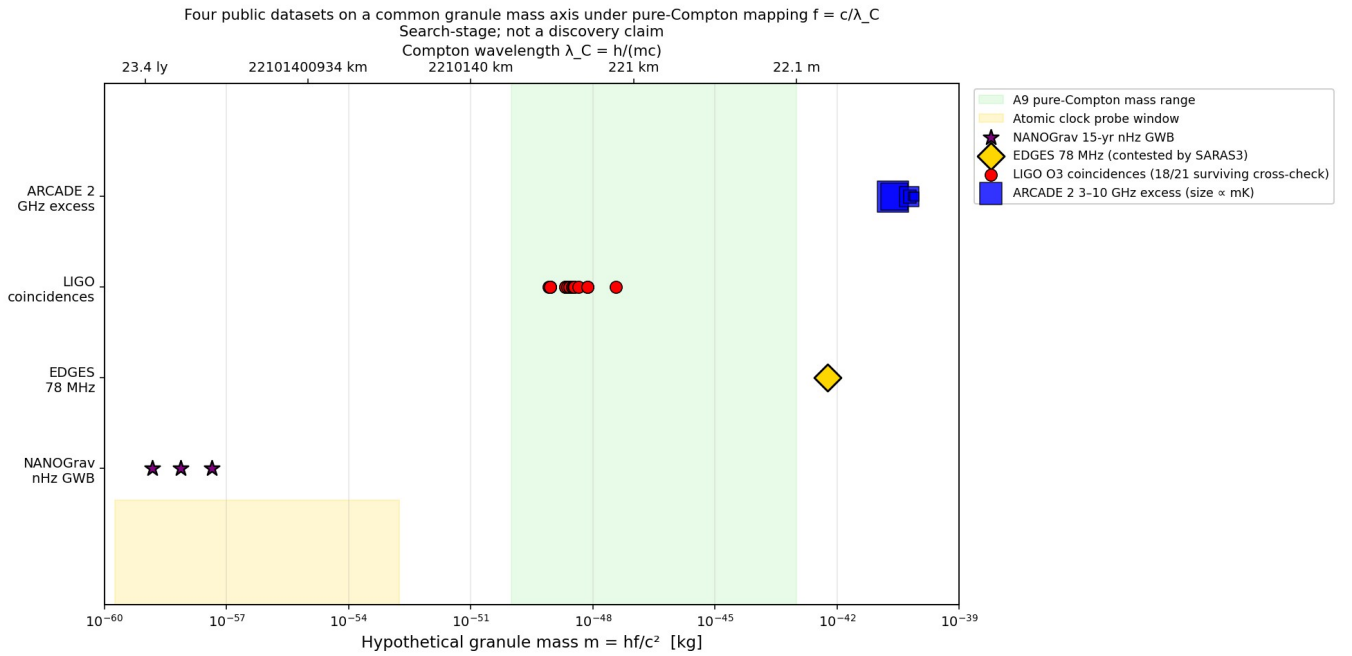


Figure 2. Four public datasets placed on a common hypothetical granule mass axis under pure-Compton mapping $f = c/\lambda_C$. Each point or band represents the central mass inferred under that single channel; the channel-ambiguity discussed above widens each band by approximately three orders of magnitude on either side. The green band marks the A9 headline range. The gold band marks the atomic-clock dark-matter probe window. The mass axis spans 21 orders of magnitude.

Two qualitative observations carry the weight of this analysis. First, both LIGO-scale and ARCADE-scale slices fall within the broad mass range that A9 predicts, while the two slices themselves are not predicted by A9 individually: the framework asks where in its wide mass spectrum existing instruments

might be sensitive, and the four datasets answer at different places consistent with the framework. Second, the structure within each slice — narrow line clustering in LIGO, smooth power-law excess in ARCADE — is informative about the local mass distribution rather than the existence of the granules themselves. A discrete family of preferred granule masses would produce LIGO-style clustering; a continuous mass distribution would produce ARCADE-style broadband excess. The combined dataset is, at this preliminary level, consistent with both being present in different parts of the spectrum.

A first scenario constraint from the cosmological constant

The observed cosmological constant has mass-equivalent energy density $\rho_\Lambda \approx 6 \times 10^{-27} \text{ kg/m}^3$ — extraordinarily dilute. (For comparison, the air in this room is 1.2 kg/m^3 , denser by twenty-seven orders of magnitude.) If granules of mass m fill space at number density n (granules per cubic metre), then the mass density contributed by them is $n \times m$. In the simplest scenario where granules are uniformly distributed and account for all of dark energy, $n \times m$ must equal ρ_Λ , giving $n = \rho_\Lambda / m$.

Granule mass slice	Mass (kg)	Required number density in the uniform-and-all scenario
NANOGrav (10 nHz)	7.4×10^{-59}	$8 \times 10^{22} / \text{mm}^3$ (effectively a continuous field)
LIGO mid (50 Hz)	3.7×10^{-49}	$1.6 \times 10^{13} / \text{mm}^3$ (dense)
EDGES (78 MHz)	5.7×10^{-43}	$1 \times 10^7 / \text{mm}^3$ (atomic-scale spacing)
ARCADE (5 GHz)	3.7×10^{-41}	$2 \times 10^5 / \text{mm}^3$ (sub-micrometre spacing)

The uniform-and-all assumption is a single simple scenario, not a forced relation. Several other possibilities deserve emphasis:

- Granules might be a fractional contribution to dark energy ($n \times m < \rho_\Lambda$), with the remainder from other sources.
- Granules might not be uniformly distributed — clustering, gravitational gradients, or mass-class-dependent density profiles are not excluded by current data.
- Different granule mass classes might have different number densities set by independent physical mechanisms, with no simple functional relation between m and n .
- A non-trivial mass-density structure could itself be a signature of new symmetries or fields — and the persistent 120-orders-of-magnitude discrepancy between the naive vacuum-energy prediction of quantum field theory and the observed cosmological constant remains the strongest hint that vacuum structure is not simple.

No granule mass slice considered here overcloses the universe at any plausible density. The table above is offered as a simple sanity-check, not a constraint of nature. A separate note on Casimir-type vacuum stresses associated with these granule masses, which addresses the relation between granule density and vacuum energy in more detail, accompanies this assessment.

Honest caveats

- All four mappings are correlational, not causal. Each dataset has independent candidate explanations from conventional physics that have not been ruled out, and at least one (EDGES) is actively contested.
- The cross-check on the LIGO clusters relies on the LSC catalogue construction methodology being correct. If LSC missed any standard resonance that happens to coincide with a cluster frequency, that cluster line should be reclassified.
- The LIGO H1 catalogue is roughly four times larger than the L1 catalogue, which complicates straightforward statistical inference; the $2.4\times$ excess at 10 mHz tolerance was computed against a local-density null, but a fuller analysis would propagate the catalogue-construction methodology.
- NANOGrav is most naturally explained by supermassive black hole binaries within standard astrophysics; the granule mapping does not displace this interpretation, it sits alongside it as one of several alternatives.
- The EDGES result is contested by SARAS 3. Should EDGES not survive replication, the EDGES point drops out of the joint figure without affecting the other three datasets.
- The ARCADE re-extraction uses a coarser foreground separation than the published work and therefore gives larger absolute excess values; the spectral index is consistent. The granule masses inferred here are robust to that difference because the mapping uses frequency only.
- The granule hypothesis itself remains a hypothesis; the framing of granule masses, clusters, or a continuous spectrum is part of the search.
- Each mapped mass is the centre of a band $\sim 10^3\text{--}10^6$ wide due to coupling-channel ambiguity, plus a measurement-precision contribution of order 10^{-4} to 1 across datasets.

Recommended next steps

- Raw-strain cross-correlation between LIGO H1 and L1 over multi-day integration periods in 2 Hz bands centred on clusters A, B, C, and E, with explicit accounting for the ~ 10 ms light-travel delay between sites. This is the decisive test for whether the catalogue-level coincidences correspond to a common physical signal.
- Extension of the same analysis to Virgo (Cascina) and KAGRA (Kamioka). A signal of physical origin should appear at the same frequencies in all four detectors with their respective light-travel delays.
- Independent replication of EDGES at 50–100 MHz by REACH, MIST, PRIZM and other cosmic-dawn radio experiments; a confirmed detection promotes the EDGES point from contested to established.
- ARCADE 2 spectrum re-extraction using the published multi-template foreground subtraction (rather than the simple latitude cut used here) to reduce residual diffuse galactic contamination and obtain the inferred granule mass with smaller systematic uncertainty; resolution of the calibration ambiguity in the 29.5, 31, 88.2, and 89.8 GHz files.
- A dedicated Sagnac / ring-laser differential experiment designed to separate Compton coupling (light-mediated) from imaginary-Kerr coupling (rotation-mediated), using two ring lasers of different perimeter in a common vault with differential analysis at 1–5 Hz. This directly probes the imaginary-

horizon channel of A4 / A7 / A8 — the only A-series coupling channel not exercised by the four datasets above.

- Bridging the mass-spectrum gap: identification of an intermediate-frequency dataset (mid-infrared, terahertz, or low-frequency radio between 100 MHz and 1 GHz) capable of probing the 10^{-48} to 10^{-42} kg gap that the four datasets above do not directly sample.
- Coordinated joint analysis of NANOGrav and LIGO bands to search for spectral features beyond the conventional astrophysics baseline that would specifically favour a granule interpretation.

Closing

Four independent public datasets — LIGO, ARCADE, EDGES, NANOGrav — collected for different purposes between 2008 and 2023, each contain unexplained features. When read through a single hypothesis from the A-series under one coupling channel, those features populate four disjoint regions of one continuous mass axis spanning twenty-one orders of magnitude. The LIGO clusters survive cross-check against the obvious instrumental and geophysical confounders. The mass inference is precise within each dataset under the assumed coupling channel; the dominant residual uncertainty is the coupling channel itself, which widens each mass slice by approximately three orders of magnitude on either side. Granule densities reproducing the observed cosmological constant are compatible across the entire mass range under the simplest uniform-and-all scenario, with significantly more freedom under realistic non-uniform scenarios.

None of this is proof. All of it is empirical reason to do the dedicated follow-up tests outlined above. The framework presented in A9 is a one-author programme conducted without institutional support over more than three decades. The consistency of this catalogue result with its predictions is offered as an empirical motivation to test those predictions properly with the resources of a dedicated team.

Note on origin. The analysis presented here developed over the course of two short workshop sessions between Claude Opus and Matey Protitch, building on Protitch's nine months of unaffiliated, unfunded work as a sole researcher. If a two-day exchange between one human and one language model can reach the level of structural assessment recorded above, a dedicated laboratory setting with a gravitational-wave-data team, a cosmology team, and a pulsar-timing team would be expected to take the work substantially further. The conditions to do so do not currently exist for this researcher.

Sources

- LIGO O3 Unidentified Spectral Lines (H1, L1) — LIGO Scientific Collaboration technical note LIGO-T2100201: <https://dcc.ligo.org/LIGO-T2100201/public>
- LIGO O3 Self-Gated Spectral Lines (calibration catalogue) — LIGO Scientific Collaboration technical note LIGO-T2100200: <https://dcc.ligo.org/LIGO-T2100200/public>
- ARCADE 2 v19 HEALPix Sky Maps — NASA LAMBDA archive: https://lambda.gsfc.nasa.gov/product/arcade/arcade_maps_get.html

- Fixsen, D. J. et al. (2011), "ARCADE 2 Measurement of the Absolute Sky Brightness at 3–90 GHz", *ApJ* 734, 5.
- Seiffert, M. et al. (2011), "Interpretation of the ARCADE 2 Absolute Sky Brightness Measurement", *ApJ* 734, 6.
- Bowman, J. D. et al. (2018), "An absorption profile centred at 78 megahertz in the sky-averaged spectrum", *Nature* 555, 67.
- Singh, S. et al. (2022), "On the detection of a cosmic dawn signal in the radio background", *Nature Astronomy* 6, 607 (SARAS 3 contesting EDGES).
- Agazie, G. et al. NANOGrav Collaboration (2023), "The NANOGrav 15-year Data Set: Evidence for a Gravitational-wave Background", *ApJL* 951, L8. arXiv:2306.16213.
- Filzinger, M. et al. (2023), "Improved Limits on the Coupling of Ultralight Bosonic Dark Matter to Photons from Optical Atomic Clock Comparisons", *Phys. Rev. Lett.* 130, 253001.
- Schumann resonance reference: Nikolaenko, A. P. & Hayakawa, M. (2002), Resonances in the Earth–Ionosphere Cavity.
- Protitch, M. ("Gravity Resonance") (2026), "The Quantum Vacuum Granule: Three Complementary Mass Scales from the 1–5 Hz Universal Hum" (A9). Zenodo DOI 10.5281/zenodo.18888582. Full A-series: <https://zenodo.org/records/18888582>